BETA TEST PLAN – Pikselite Engine

1. Project Context

Pikselite Engine is a 2D pixel-based game engine focused on fine-grained pixel manipulation and simulation. It allows users to create projects, design sprites at the pixel level, using elements like water and fire, and simulate their interactions in real time.

The engine provides:

- A sprite editor with pixel Elements
- A game editor to create the game
- A Scripting-based event system
- A real-time preview mode
- A build system to generate playable executables

The goal of this beta test is to validate the stability, usability, and correctness of the core editor workflows before public release.

Definitions:

- Game Object: Entity in the game that can be linked using component.
- Sprite: Graphical element that can be moved in the renderer.
- PNG sprite: Graphical element using PNG images.
- Pixel sprite: Graphical element composed of pixel elements.

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2. User Roles

The following roles will be involved in beta testing.

Role Name	Description
Game Editor	Uses the engine to create projects, sprites, scenes, and gameplay logic
Player	Launches the built game and plays it

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3. Feature Table

The following features will be shown and validated during the defense.

Feature ID	User Role	Feature Name	Short Description
F1	Game Editor	Project Management	Create a new project

Feature ID	User Role	Feature Name	Short Description
F2	Game Editor	Project Management	Save a project
F3	Game Editor	Project Management	Open an existing project
F4	Game Editor	Project Management	Import a project
F5	Game Editor	Pixel sprite Creation	Create Pixel sprites using the sprite editor
F6	Game Editor	Pixel sprite Creation	Edit Pixel sprite using the sprite editor
F7	Game Editor	Pixel sprite Creation	Save Pixel sprite using the sprite editor
F8	Game Editor	Pixel sprite Creation	Load Pixel sprite using the sprite editor
F9	Game Editor	Pixel sprite Creation	Remove pixels using the eraser
F10	Game Editor	Pixel sprite Creation	Switch pixel element brush
F11	Game Editor	Pixel sprite Creation	Resize the brush
F12	Game Editor	Pixel sprite Creation	Pixel sprite color pattern
F13	Game Editor	Pixel Elements	Place Stone element
F14	Game Editor	Pixel Elements	Place Dirt element
F15	Game Editor	Pixel Elements	Place Wood element
F16	Game Editor	Pixel Elements	Place Sand element
F17	Game Editor	Pixel Elements	Place Water element
F18	Game Editor	Pixel Elements	Place Lava element
F19	Game Editor	Pixel Elements	Place Fire element
F20	Game Editor	Game Editing	Create PNG sprite game object to current scene
F21	Game Editor	Game Editing	Create Pixel sprite game object to current scene
F22	Game Editor	Game Editing	Rename a game object
F23	Game Editor	Game Editing	Remove a game object
F24	Game Editor	Game Editing	Move a game object
F25	Game Editor	Game Editing	Enable/Disable a game object
F26	Game Editor	Game Editing	Add a physics component to an existing game object

Feature ID	User Role	Feature Name	Short Description
F27	Game Editor	Game Editing	Add gravity force to a physics game object
F28	Game Editor	Game Editing	Add rotation locker to the a physics game object
F29	Game Editor	Game Editing	Remove a physics component to an existing game object
F30	Game Editor	Game Editing	Add a velocity component to an existing game object
F31	Game Editor	Game Editing	Remove a velocity component to an existing PNG sprite game object
F32	Game Editor	Game Editing	Add a sprite component to an existing game object
F33	Game Editor	Game Editing	Resize a sprite game object using sprite component
F34	Game Editor	Game Editing	Change renderer layer of a sprite game object
F35	Game Editor	Game Editing	Remove a sprite component to an existing game object
F36	Game Editor	Game Editing	Add a script component to an existing game object
F37	Game Editor	Game Editing	Remove a script component to an existing game object
F38	Game Editor	Game Editing	Configure sprite game object events using scripting
F39	Game Editor	Asset Management	Manage scenes via asset explorer
F40	Game Editor	Game Preview	Simulate gameplay in real time inside the editor
F41	Game Editor	Pixel Simulation	Simulate pixel physics
F42	Game Editor	Pixel Simulation	Simulate pixel interactions

Feature ID	User Role	Feature Name	Short Description
F43	Game Editor	Pixel Simulation	Simulate particle interaction with pixels
F44	Game Editor	Scene Management	Create a scene
F45	Game Editor	Scene Management	Remove a scene
F46	Game Editor	Scene Management	Load a scene in project editor
F47	Game Editor	Scene Management	Reload scene in build/preview
F48	Game Editor	Scene Management	Switch scene in build/preview
F49	Game Editor / Player	Game Build	Generate a playable executable

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4. Success Criteria

Feature ID	Key Success Criteria	Indicator / Metric	Result
F1	Projects can be created	No errors, correct folder structure generated	Done
F2	Projects can be saved	Changes are correctly saved without corruption	Done
F3	Existing projects can be opened	Project loads successfully with all assets restored	Done
F4	Projects can be imported from another folder	Imported project structure and assets are preserved	Done
F5	Pixel sprite can be created	Pixel sprite appears correctly in sprite editor	Done
F6	Pixel sprite can be edited	Pixel modifications are applied correctly	Done
F7	Pixel sprite can be saved	Pixel sprite file is correctly stored and reusable	Done
F8	Pixel Sprite can be loaded	Sprite renders correctly with preserved Elements	Done
F9	Pixel Sprite has color pattern on pixels	Pixels of the same element have different	Done

Feature ID	Key Success Criteria	Indicator / Metric	Result
		colors when drawn using a color palette	
F10	Pixels can be erased using the eraser tool	Selected pixels are removed correctly	Done
F11	Pixel element brush can be switched	The brush draws the element choosen	Done
F12	Brush size can be resized	Brush size updates correctly and affects drawing area	Done
F13	Stone elements can be placed	Stone pixels are created and behave as solid elements	Done
F14	Dirt elements can be placed	Dirt pixels are created and rendered correctly	Done
F15	Wood elements can be placed	Wood pixels are created and support flammable interactions	Done
F16	Sand elements can be placed	Sand pixels fall and accumulate correctly	Done
F17	Water elements can be placed	Water pixels flow correctly during simulation	Done
F18	Lava elements can be placed	Lava interacts correctly with surrounding elements	Done
F19	Fire elements can be placed	Fire spreads and disappears according to simulation rules	Done
F20	PNG prite game objects can be created in the current scene	Game object appears correctly in scene and viewport	Done
F21	Pixel sprite game objects can be created in the current scene	Pixel object is added correctly to the scene	Done
F22	Game objects can be renamed	New object name is updated and persisted correctly	Done
F23	Game objects can be removed	Object disappears from scene without errors	Done
F24	Game objects can be moved	Object position updates correctly in scene	Done
F25	Game objects can be enabled/disabled	Objects are correctly Enabled/disabled in the	Done

Feature ID	Key Success Criteria	Indicator / Metric	Result
		preview so they are hidden and cannot interact with other objects.	
F26	Physics components can be added to objects	Physics behavior is enabled correctly	Done
F27	Gravity force can be added to physics objects	Object reacts correctly to gravity simulation	Done
F28	Rotation lock can be added to physics objects	Object rotation remains constrained during physics simulation	Done
F29	Physics components can be removed	Object no longer reacts to physics simulation	Done
F30	Velocity components can be added	Game Object movement velocity is applied correctly	Done
F31	Velocity components can be removed	Object stops using velocity-based movement	Done
F32	Sprite components can be added	Sprite renders correctly on the game object	Done
F33	Sprite game objects can be resized	Sprite scaling updates visually and physically correctly	Done
F34	Renderer layers can be changed	Rendering order updates correctly in scene	Done
F35	Sprite components can be removed	Sprite no longer renders on the object	Done
F36	Script components can be added	Scripts are attached and initialized correctly	Done
F37	Script components can be removed	Scripts no longer affect the game object	Done
F38	Sprite Game object events can be configured using scripting	Scripted events trigger and execute correctly	Done
F39	Scenes can be managed in the asset explorer	Scenes can be created, renamed, moved, and deleted correctly	Done
F40	Real-time preview can be launched inside the editor	Preview runs stably and reflects current scene state	Done

Feature ID	Key Success Criteria	Indicator / Metric	Result
F41	Pixel physics simulation behaves correctly	Pixel movement and physics interactions are accurate	Done
F42	Pixel interactions behave correctly	Elements interact according to defined simulation rules	Done
F43	Simulate particles interaction with pixels correctly	Particle are render with float velocity and can transform into pixels if possible	Done
F44	Scenes can be created	New scene file appears in assets and opens without errors	Done
F45	Scenes can be removed	Scene file is deleted and no longer appears in explorer	Done
F46	Scenes can be loaded in project editor	Selected scene loads with its objects and settings intact	Done
F47	Scenes can be reloaded in build/preview	Preview resets to the saved scene state without errors	Done
F48	Scenes can be switched in build/preview	Active scene changes correctly and renders the new content	Done
F49	Playable executables can be generated	Build completes successfully and generated game launches correctly	Done